

SOUTH EASTERN UNIVERSITY OF SRI LANKA

SWT22022 Practical for Internet Application Development

Proposal for CA1 Group Assignment

Project Title	BusLink.lk
Group ID	Group M
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1. Project Overview

BusLink.lk is a web-based system built to bring some order to how private bus services in Sri Lanka are run and used. Right now, bus owners, drivers, conductors, passengers, and administrators all operate somewhat separately. This platform puts them on one system, covering buses, routes, schedules, trips, bookings, and ticketing.

Bus owners get a central place to manage their fleet, staff, routes, and schedules instead of juggling separate records. Passengers can search for buses, check routes and timings, book a seat, get a digital ticket, and follow their bus on a map as it approaches. The platform also handles basic communication, so operators can push out delay notices or service updates directly to passengers.

The end goal is straightforward: cut down the manual work bus owners deal with day to day, and give passengers something more reliable than guesswork when it comes to bus timing and location.

2. Problem Statement

Private buses carry a large share of daily commuters in Sri Lanka, yet most operators still run things on paper or by memory. tracking buses, staff, routes, and tickets without any real digital system behind it. That makes it hard to keep records straight or get a clear picture of how operations are actually going.

Passengers deal with a parallel problem. There's rarely a dependable way to check a schedule, confirm a bus is actually running, or find out where it currently is. When a bus is late, passengers usually find out by waiting at the stop and watching the road, not from any notification. Most of what they know comes from posted timetables, word of mouth, or habit.

Neither side has a proper digital system to fall back on. Owners need better tools to run their operations, and passengers need a way to get accurate, timely information without relying on guesswork. That gap is what BusLink.lk is meant to close.

3. Proposed Solution

BusLink LK is a single web platform connecting bus owners, bus staff, passengers, and administrators. Owners get structured tools to manage their buses, staff, routes, schedules, and trips in one place, rather than spreading records across notebooks or spreadsheets. Passengers get a simple interface to find and book the buses they need.

On the passenger side, users can search for buses matching their route and timing, check schedules, book trips, and receive a digital ticket instead of a paper one. Using the GPS on a passenger's phone, the system also tracks the booked bus in real time and sends a notification once it's close to the pickup point.

For owners and staff, the platform covers the everyday operational side. bookings, ticketing, payments, and service updates. so less of it has to be tracked manually. Administrators oversee the platform itself, managing users and handling any issues that come up. Put together, these pieces are meant to reduce manual overhead, improve communication between operators and passengers, and make private bus travel a bit more predictable for everyone involved.

4. Project Objectives

Main Objective

To develop a centralized web-based platform that improves how private bus services are managed in Sri Lanka, while giving passengers a reliable and convenient way to access bus information and transportation services.

Specific Objectives

I. To build a centralized management system that lets private bus owners handle buses, drivers, conductors, routes, schedules, and trips from one place, rather than managing these separately.

II. To give passengers a straightforward way to check bus information, including routes, schedules, available services, and pickup and drop-off points.

III. To set up an online booking and digital ticketing system, so passengers can book a trip and receive a digital ticket that can be verified through a QR code.

IV. To add real-time bus tracking using the location services on a passenger's phone, so they can see where their booked bus currently is.

V. To send location-based notifications that alert passengers once their booked bus is approaching their selected pickup point.

VI. To handle digital payments and keep records of passenger transactions, ticket sales, and trip collections for each bus.

VII. To include reporting tools that help bus owners track bookings, revenue, bus performance, and other operational details.

VIII. To give administrators the tools to manage users, bus services, and oversee overall platform activity.

IX. To build the system as a full-stack web application using appropriate technologies, with the development process tracked through version control from start to finish.

5. Target Users / Stakeholders

User / Stakeholder	Role / Description	Main Needs
Bus Owner	Manages private buses and their operations through the system.	Manage buses and bus staff, assign buses to routes, manage schedules and trips, and view bookings and payments.
Bus Staff	Includes drivers and conductors who are responsible for operating and managing an assigned bus trip.	View assigned bus and route details, manage trip activities, update trip status, verify tickets, and share the bus location.
Passenger	Uses the system to find and use available private bus services.	View routes and schedules, find suitable buses, book trips, receive digital tickets, make payments, track buses, and receive notifications.
System Administrator	Oversees the overall system and manages platform-related activities.	Manage users, verify bus owners and buses and monitor the system.

6. Proposed Technology Stack

Layer / Technology	Proposed Choice & Brief Justification
Frontend	**React.js + Vite + Tailwind CSS** - Used to develop a responsive and user-friendly web interface for passengers, bus owners, bus staff, and administrators.
Backend / API	**Node.js + Express.js + Socket.IO** - Node.js and Express.js will provide the backend and REST APIs, while Socket.IO will support real-time bus location and trip-status updates.
Database	**MongoDB + Mongoose** - Used to store and manage users, buses, bus staff, routes, schedules, trips, bookings, tickets, payments, and location data. Mongoose will provide structured data models and validation.
Authentication / Security	**JWT + bcrypt + Role-Based Access Control** - JWT will handle authentication, bcrypt will securely hash passwords, and role-based access control will restrict system functions according to user roles.
Other Tools / Libraries	**Leaflet + OpenStreetMap + Geolocation API + OSRM + QR Code Library + Axios + Git/GitHub** - Used for maps, real-time location tracking, route and distance information, digital ticket QR codes, frontend API communication, and version control.

7. Major Functional Modules

Module	Description / Main Functions	Primary Member	Expected Complexity / Contribution
Module 1 - Bus & Route Management	Manage buses, bus staff, routes, bus stops, and schedules. Bus owners can register buses, add bus staff, assign buses to routes, and create or update schedules.	M.P.S.A. Nawanjana	High - Includes multiple related entities, validation, route assignment, schedule management, and backend business rules.
Module 2 - Trip, Booking & Digital Ticketing	Manage scheduled trips and passenger bookings. Passengers can select a trip, pickup and drop-off locations, and receive a digital ticket with a QR code. Bus staff can verify tickets during a trip.	K.L.H.G. Karandana	High - Includes booking logic, trip capacity, ticket generation, QR verification, and booking status management.
Module 3 - Real-Time Bus Tracking & Notifications	Track the live location of buses using the bus staff member's mobile device. Passengers can view their bus on a map and receive a notification when the bus reaches the defined distance from their pickup location.	W.A.S.S. Kaluwila	Very High - Involves browser geolocation, maps, real-time communication, distance calculation, location updates, and notification logic.
Module 4 - Payment & Revenue Management	Manage passenger payments and transaction records related to bookings. Bus owners can view payment information, trip collections, and revenue summaries.	W.M.S.R. Weerasinghe	High - Includes payment processing, transaction records, booking-payment relationships, validation, and revenue calculations.
Module 5 - User, Reports & Administration Management	Manage system users and administrative activities. Administrators can manage users, verify bus-related information, handle complaints and reports, while passengers can submit ratings and feedback about their bus service.	B.V.K. Sathsara	Medium - Includes role-based access, administrative workflows, complaint handling, ratings, and user management.

8. Individual Contribution Plan

Member / Reg. No.	Assigned Module / Responsibility	Key Tasks	Expected Deliverables
M.P.S.A. Nawanjana	Module 1 – Bus & Route Management / Frontend	Design and develop the bus owner interfaces. Implement interfaces for bus registration, bus staff management, route and bus-stop management, bus assignment to routes, and schedule management. Connect the frontend with relevant backend APIs and handle client-side validation.	Bus owner dashboard, bus management pages, bus staff pages, route and bus-stop interfaces, schedule management interfaces, responsive UI, and API integration.
K.L.H.G. Karandana	Module 2 – Trip, Booking & Digital Ticketing / Frontend	Design and develop passenger-facing interfaces for viewing routes and schedules, selecting trips, choosing pickup/drop-off locations, making bookings, and viewing booking history. Implement the digital ticket interface and QR-code display. Integrate the frontend with booking and ticket APIs.	Passenger dashboard, trip and booking interfaces, booking workflow, booking history, digital ticket page, QR-code display, responsive UI, and API integration.
W.A.S.S. Kaluwila	Module 3 – Real-Time Bus Tracking & Notifications / Backend	Develop APIs and backend services for bus trip operations and live location tracking. Process location data received from bus staff devices, calculate distances from passenger pickup locations, manage real-time updates, and implement bus-approaching notifications.	Trip and tracking APIs, location processing services, real-time communication, distance/geofence logic, notification services, and related database models.
W.M.S.R. Weerasinghe	Module 4 – Payment & Revenue Management / Backend	Develop backend services for payment processing and transaction management. Connect payments with passenger bookings and tickets, maintain transaction records, validate payment status, and calculate	Payment APIs, transaction database models, payment validation, booking-payment integration, revenue calculation services, APIs.

		revenue and trip collection summaries for bus owners.	
B.V.K. Sathsara	Module 5 – User, Reports & Administration Management / Backend	Develop authentication and authorization services, user and role management, administrative functions, bus-owner verification, complaint and report handling, ratings, and feedback management. Implement security and access control for different user roles.	Authentication APIs, role-based access control, user management APIs, administration services, complaint/report management, services, and security implementation.

9. Key Functional Requirements

I. User Registration and Authentication - The system shall allow users to register and securely log in according to their assigned role, including bus owners, bus staff, passengers, and administrators.

II. Bus Management - The system shall allow bus owners to register, update, and manage their buses and relevant bus information.

III. Bus Staff Management - The system shall allow bus owners to add and manage bus staff and assign staff members to their respective buses and trips.

IV. Route and Bus Stop Management - The system shall allow bus owners to create and manage routes containing multiple bus stops and assign buses to available routes.

V. Schedule Management - The system shall allow bus owners to create and update bus schedules, including departure times, route information, and assigned buses.

VI. Bus and Schedule Search - The system shall allow passengers to search and view available buses, routes, stops, and schedules based on their travel requirements.

VII. Passenger Booking - The system shall allow passengers to select an available trip, specify pickup and drop-off locations, and make a booking.

VIII. Digital Ticket Generation and Verification - The system shall generate a digital ticket for a successful booking and provide a QR code that bus staff can use to verify the ticket.

IX. Payment Management - The system shall record and manage payments associated with passenger bookings and maintain transaction information.

X. Revenue Management - The system shall allow bus owners to view booking-related payments, trip collections, and revenue summaries.

XI. Real-Time Bus Location Tracking - The system shall obtain the current location of an assigned bus through the bus staff member's mobile device and provide updated bus locations to authorized users.

XII. Map-Based Bus Tracking - The system shall display the current bus location and relevant route information on an interactive map for passengers.

XIII. Bus-Approaching Notification - The system shall calculate the distance between a bus and a passenger's selected pickup location and notify the passenger when the bus reaches the configured distance.

XIV. Complaint and Report Management - The system shall allow passengers to submit complaints or reports and allow administrators to review and manage them.

XV. Administrative Management - The system shall allow administrators to manage users, review bus-related information, handle complaints and reports, and monitor important system activities.

XVI. Role-Based Access Control - The system shall restrict access to system functions and information according to the user's role and permissions.

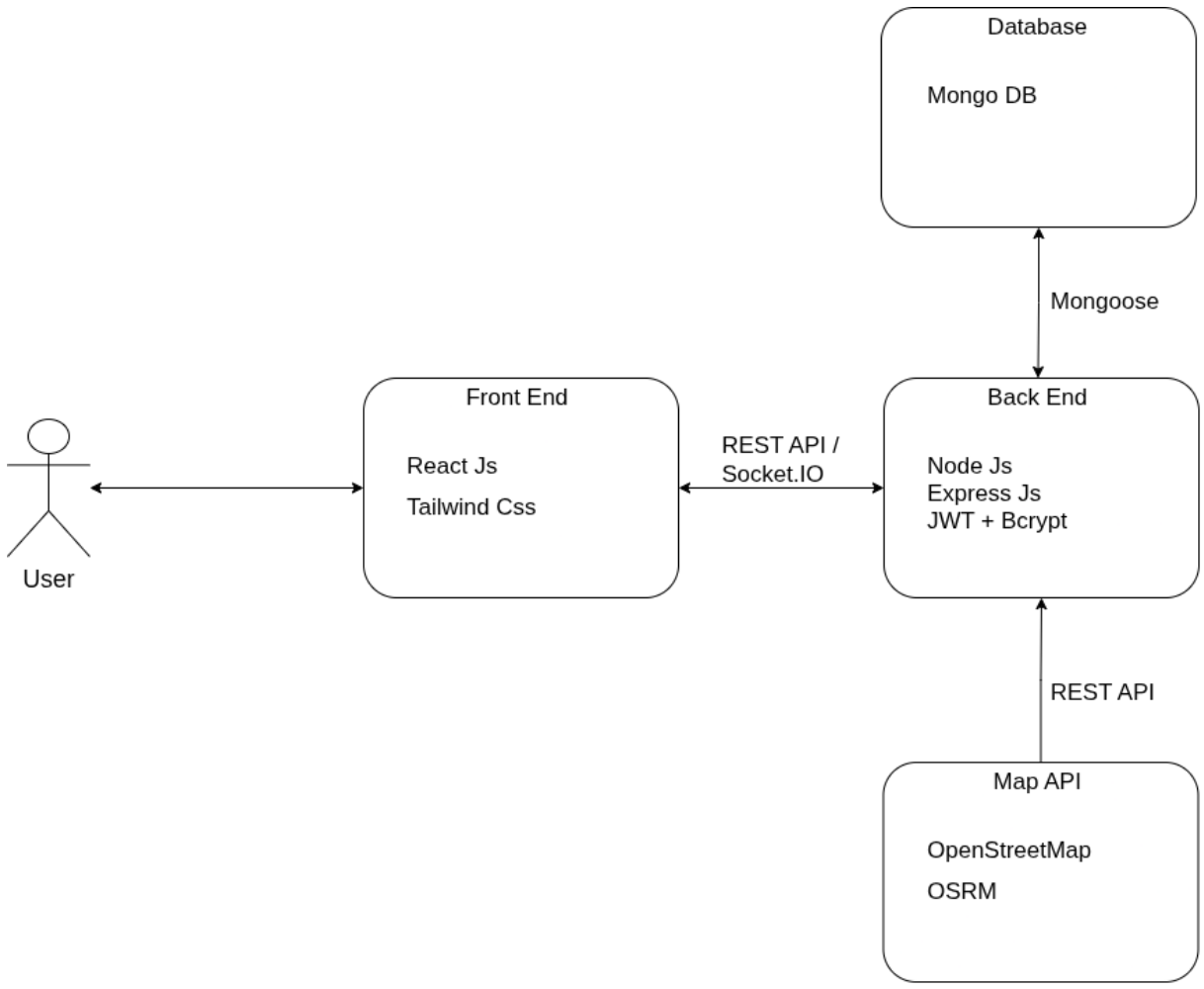
XVII. Notification Management - The system shall provide relevant notifications/emails to users regarding bookings, payments, trip updates, bus location events, and other important service information.

10. High-Level System Architecture

The proposed system will use a three-layer web application architecture consisting of the frontend, backend, and database. The React.js frontend will provide interfaces for passengers, bus owners, bus staff, and administrators, while Tailwind CSS will be used for responsive interface design.

The Node.js and Express.js backend will handle the main application logic and provide APIs for communication between the frontend and database. JWT and bcrypt will support authentication and security. The backend will also manage real-time bus location and other system operations.

MongoDB will store the application's main data, including users, buses, bus staff, routes, schedules, trips, bookings, tickets, and payments. The Map API services (OpenStreetMap and OSRM) will support map display and route-related functions such as displaying bus routes and calculating distances.



11. Preliminary Database Design

If applicable, provide an initial list of the main entities/collections/tables and their purpose. A detailed database design is not required at proposal stage.

Entity / Collection / Table	Purpose	Important Data
Users	Stores all registered users and their roles.	Name, email, password, role, contact details, account status
Buses	Stores information about private buses managed by bus owners.	Registration number, owner ID, bus details, capacity, status
Bus Staff	Stores drivers and conductors assigned to buses.	Name, contact details, role, assigned bus, status
Routes	Stores bus routes created and managed by bus owners.	Route name, owner ID, start point, destination, assigned bus
Bus Stops	Stores stops belonging to a route.	Stop name, route ID, location, stop order
Schedules	Stores planned operating times for buses on routes.	Bus ID, route ID, departure time, arrival time, operating days
Trips	Represents an individual bus journey based on a schedule.	Bus ID, route ID, staff, date, departure time, trip status
Bookings	Stores passenger bookings for specific trips.	Passenger ID, trip ID, pickup stop, drop-off stop, booking status
Tickets	Stores digital tickets generated from successful bookings.	Ticket ID, booking ID, QR code, fare, ticket status
Payments	Stores payment and transaction information related to bookings.	Booking ID, amount, payment method, transaction ID, payment status
Bus Locations	Stores recent GPS location information for buses during active trips.	Bus ID, trip ID, latitude, longitude, timestamp
Notifications	Stores notifications sent to users about bookings, trips, and bus locations.	User ID, message, notification type, status, timestamp
Feedback & Complaints	Stores passenger ratings, feedback, and complaints about bus services.	Passenger ID, trip ID, rating, feedback, complaint details, status

12. Proposed Development Plan

Provide a realistic plan for developing and completing the project (Gantt chart). You may adapt the weeks to the course schedule.



13. Expected Outcome

The completed BusLink LK system is expected to provide a centralized platform for managing private bus services and accessing passenger transportation services. Bus owners will be able to manage their buses, bus staff, routes, schedules, trips, bookings, and revenue in an organized manner. Passengers will be able to find suitable bus services, book trips, receive digital tickets, make payments, track buses, and receive important trip notifications.

The system is expected to reduce manual work, improve access to reliable bus information, and improve communication between passengers and private bus service providers. Overall, BusLink LK aims to provide a more convenient and organized experience for both private bus operators and passengers.

14. References / Supporting Sources

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| 2 | Kumarage, A. S., & Jayaratne, M. D. R. P. (2008). *Lessons in ownership, regulation and management from 100 years of bus transport in Sri Lanka*. Research in Transportation Economics. URL: https://doi.org/10.1016/j.retrec.2008.05.025 | Section 1 – Project Overview and Section 2 – Problem Statement. Used to understand the management and service-quality challenges of Sri Lanka's bus transportation sector. ([ScienceDirect][2]) |

| 3 | Central Provincial Department of Transport (2026). *Developing the attitude of drivers and conductors engaged in private bus services*. URL: https://dot.cp.gov.lk/developing-the-attitude-of-drivers-and-conductors-engaged-in-private-bus-services/ | Section 2 – Problem Statement and Section 5 – Target Users / Stakeholders. Supports the importance of drivers, conductors, passenger service, operational issues, and staff responsibilities in private bus transportation. ([Department of Posts][3]) |

| 4 | Samarasinghe, S. A. S. V., et al. (2025). *A GPS-based Real-time Bus Tracking System for Improving Commuter Experience in Sri Lanka's Public Transportation*. ICIET 2025. URL: https://iciet.sjp.ac.lk/wp-content/uploads/2025/12/ICIET2025_Proceeding__F005_Final_251230.pdf | Sections 2, 3, 4, 7, 9 & 13. Used to understand the need for real-time bus location, schedule information, and improved passenger experience through GPS-based tracking. ([ICIEt SJP][4]) |

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| 6 | *Mobile Application for Real-Time Bus Tracking and Seat Booking*. Open University of Sri Lanka Research Sessions. URL: https://ours.ou.ac.lk/wp-content/uploads/2021/10/ID-16_MOBILE-APPLICATION-FOR-REAL-TIME-BUS-TRACKING-AND-SEAT-BOOKING.pdf | Sections 2, 3, 4, 7 & 9. Used to understand problems related to bus tracking, passenger reservations, seat booking, and communication between passengers and bus operators. ([OU Services][6]) |

| 7 | Brakewood, C., & Watkins, K. (2019). *A literature review of the passenger benefits of real-time transit information*. Transport Reviews. URL: https://doi.org/10.1080/01441647.2018.1472147 | Sections 2, 3, 4 & 13. Used to support the value of providing passengers with real-time bus information and arrival information. ([Taylor & Francis Online][7]) |

| 8 | Tang, L., & Thakuriah, P. (2012). *Ridership effects of real-time bus information system: A case study in the City of Chicago*. Transportation Research Part C. URL: https://doi.org/10.1016/j.trc.2012.01.001 | Sections 3, 4 & 13. Used as supporting evidence for the potential benefits of real-time bus information for passengers and bus services. ([ScienceDirect][8]) |

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